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AQA A Level Further Mathematics (7367)

Topic D: Further Algebra and Functions

Sub topic D14: Stationary Points Without Calculus

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic	**D: Further Algebra and Functions**
Sub topic	**D14: Stationary Points Without Calculus**

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To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

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Name : ____________________

Date : ____________________

Time : ____________________

Total Marks Available : _____

Total Marks Received : _____

Q1.

A curve C_1 has equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = \dfrac{5x^2 - 12x + 12}{x^2 + 4x - 4}$

The line $y = k$ intersects the curve C_1

(a) (i) Show that $(k + 3)(k - 1) > 0$

(5)

(ii) Hence find the coordinates of the stationary point of C_1 that is a maximum point.

(4)

(b) Show that the curve C_2, whose equation is $y = \dfrac{1}{\mathrm{f}(x)}$, has no vertical asymptotes.

(2)

(c) State the equation of the line that is a tangent to both C_1 and C_2.

(1)

(Total 12 marks)

Q2.

A curve has equation

$$y = \frac{x^2 - 2x + 1}{x^2 - 2x - 3}$$

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(a) Find the equations of the three asymptotes of the curve.

(3)

(b) (i) Show that if the line $y = k$ intersects the curve then

$$(k - 1)x^2 - 2(k - 1)x - (3k + 1) = 0$$

(1)

(ii) Given that the equation $(k - 1)x^2 - 2(k - 1)x - (3k + 1) = 0$ has real roots, show that

$$k^2 - k \geq 0$$

(3)

(iii) Hence show that the curve has only one stationary point and find its coordinates.

(No credit will be given for solutions based on differentiation.)

(4)

(c) Sketch the curve and its asymptotes.

(3)

(Total 14 marks)

Q3.

A graph has equation

$$y = \frac{x-4}{x^2+9}$$

(a) Explain why the graph has no vertical asymptote and give the equation of the horizontal asymptote.

(2)

(b) Show that, if the line $y = k$ intersects the graph, the x-coordinates of the points of intersection of the line with the graph must satisfy the equation

$$kx^2 - x + (9k + 4) = 0$$

(2)

(c) Show that this equation has real roots if $-\frac{1}{2} \leq k \leq \frac{1}{18}$.

(5)

(d) Hence find the coordinates of the two stationary points on the curve.

(No credit will be given for methods involving differentiation.)

(6)

(Total 15 marks)

Q4.

A curve has equation

$$y = \frac{x^2}{(x-1)(x-5)}$$

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(a) Write down the equations of the three asymptotes to the curve.

(3)

(b) Show that the curve has no point of intersection with the line $y = -1$.

(3)

(c) (i) Show that, if the curve intersects the line $y = k$, then the x-coordinates of the points of intersection must satisfy the equation

$$(k - 1)x^2 - 6kx + 5k = 0$$

(2)

(ii) Show that, if this equation has equal roots, then

$$k(4k + 5) = 0$$

(2)

(d) Hence find the coordinates of the two stationary points on the curve.

(5)

(Total 15 marks)

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Q5.

A curve C has equation

$$y = \frac{2}{x(x-4)}$$

(a) Write down the equations of the three asymptotes of C.

(3)

(b) The curve C has one stationary point. By considering an appropriate quadratic equation, find the coordinates of this stationary point.

(No credit will be given for solutions based on differentiation.)

(6)

(c) Sketch the curve C.

(3)

(Total 12 marks)

Q6.

A curve C has equation

$$y = \frac{(x+1)(x-3)}{x(x-2)}$$

(a) (i) Write down the coordinates of the points where C intersects the x-axis.

(2)

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(ii) Write down the equations of all the asymptotes of C.

(3)

(b) (i) Show that, if the line $y = k$ intersects C, then

$$(k-1)(k-4 \geq 0$$

(5)

(ii) Given that there is only one stationary point on C, find the coordinates of this stationary point.

(No credit will be given for solutions based on differentiation.)

(3)

(c) Sketch the curve C.

(3)

(Total 16 marks)

Q7.

The function f is defined by

$$f(x) = \frac{x^2 + 2x + 2}{x^2}$$

(a) Write down the equations of the two asymptotes to the curve $y = f(x)$.

(2)

(b) By considering the expression $x^2 + 2x + 2$:

(i) show that the graph of $y = f(x)$ does not intersect the x-axis;

(2)

(ii) find the non-real roots of the equation $f(x) = 0$.

(3)

(c) (i) Show that, if the equation $f(x) = k$ has two equal roots, then

$$4 - 8(1 - k) = 0$$

(3)

(ii) Deduce that the graph of $y = f(x)$ has exactly one stationary point and find its coordinates.

(4)

(Total 14 marks)

Q8.

The function f is defined by

$$f(x) = \frac{x^2 + 4x}{x^2 + 9}$$

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(a) (i) The graph of $y = f(x)$ has an asymptote which is parallel to the x-axis. Find the equation of this asymptote.

(1)

(ii) Explain why the graph of $y = f(x)$ has no asymptotes parallel to the y-axis.

(2)

(b) Show that the equation $f(x) = k$ has two equal roots if $9k^2 - 9k - 4 = 0$.

(4)

(c) Hence find the coordinates of the two stationary points on the graph of $y = f(x)$.

(6)

(Total 13 marks)